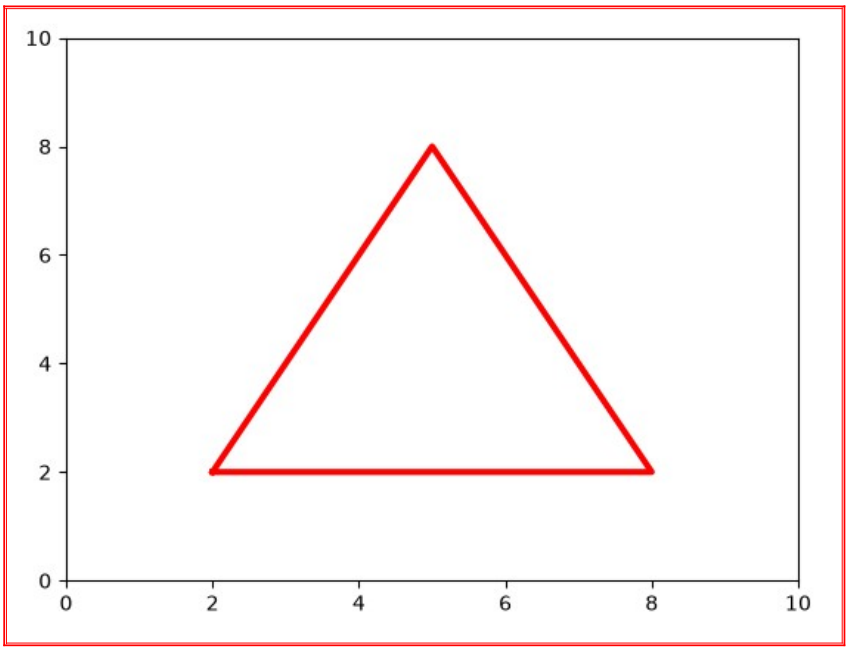
TD 2 - Matplotlib

Exercice 1 :

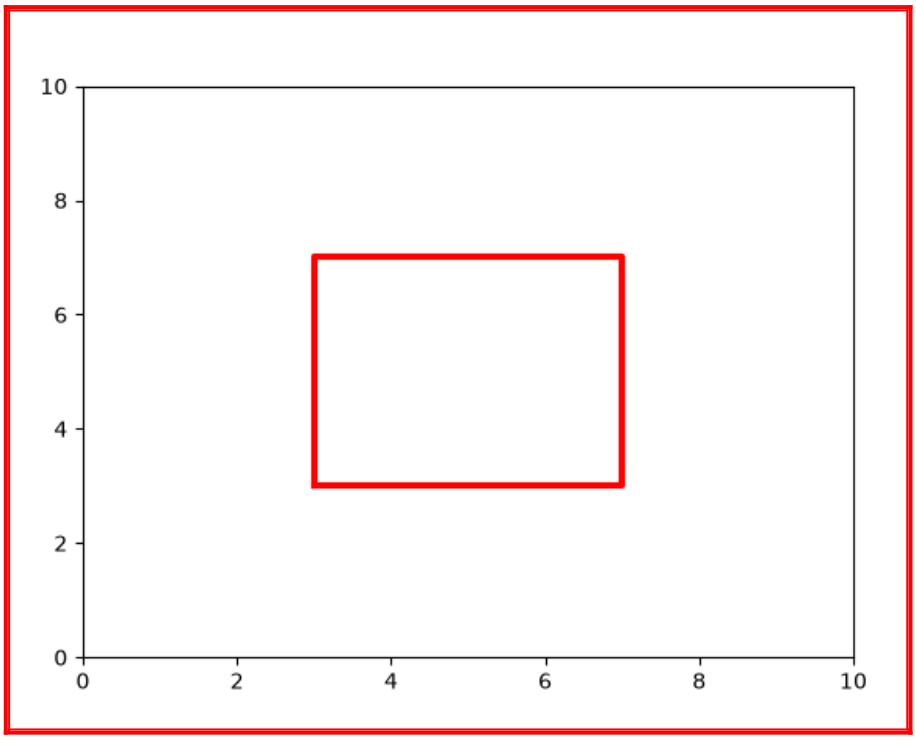
```
from pylab import *
xlim(0,10)
ylim(0,10)
x=[2,5,8,2]
y=[2,8,2,2]
plot(x,y,linewidth=3,*'r')
show()
```



Le triangle est bien forme.

Exercice 2 :

```
from pylab import *
xlim(0,10)
ylim(0,10)
x=[3,7,7,3,3]
y=[3,3,7,7,3]
plot(x,y,linewidth=3,*'r')
show()
```



Le carré est bien formé.

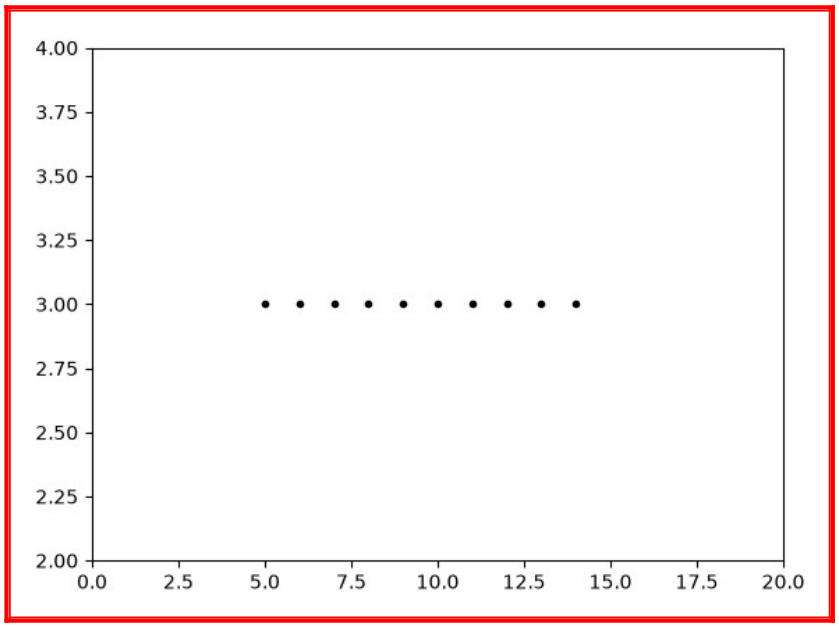
Exercice 3 :

```
from pylab import *

listex = []
listey = []

for x in range(5, 15):
  listex.append(x)
  listey.append(3)

plot(listex, listey, '.k')
xlim(0, 20)
ylim(2.0, 4.0)
show()
```



Exercice 4 :

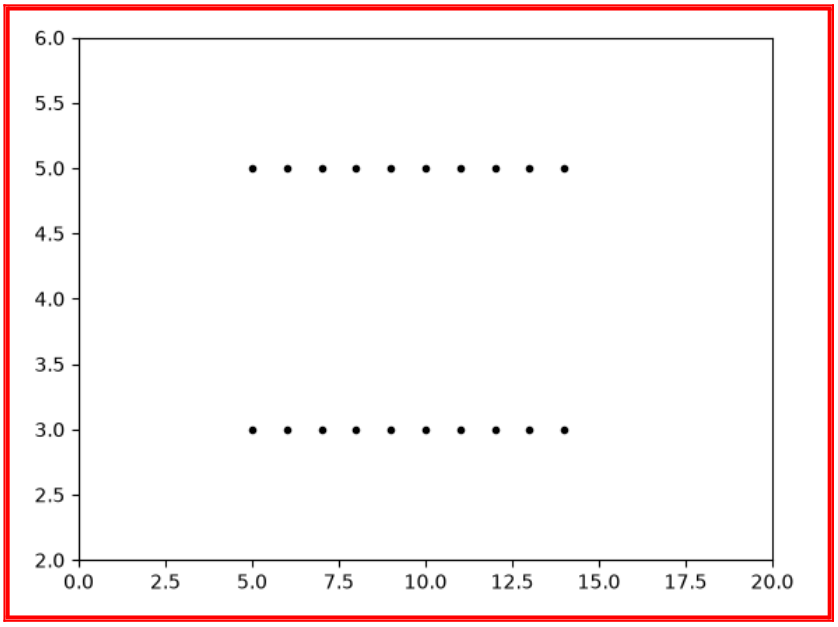
```
from pylab import *

listex = []
liste_y1 = []
liste_y2 = []

for x in range(5, 15):
  listex.append(x)
  liste_y1.append(3)
  liste_y2.append(5)

plot(listex, liste_y1, '.k')
plot(listex, liste_y2, '.k')
xlim(0, 20)
ylim(2.0, 6.0)
show()
```

Forme :



Exercice 5 :

```
from pylab import *

listex = []
listey = []

for y in range(10, 20):

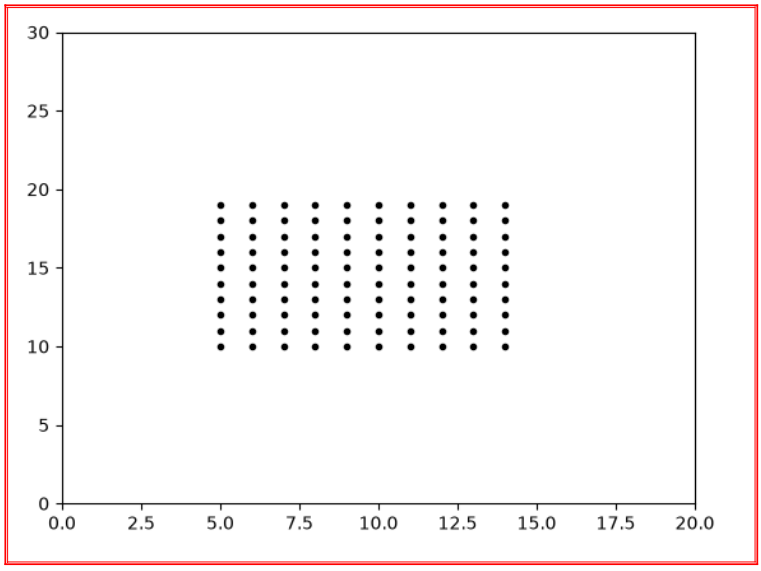
  for x in range(5, 15):
    listex.append(x)
    listey.append(y)

plot(listex, listey, '.k')

xlim(0, 20)
ylim(0, 30)

show()
```

Forme :



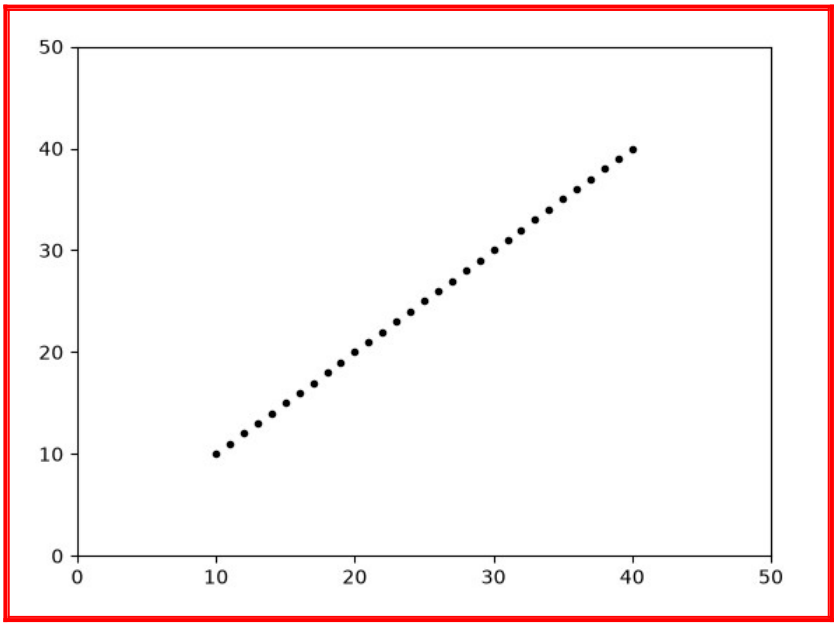
Exercice 6 :

```
from pylab import *

listex = []
listey = []

for x in range(10, 41):
  listex.append(x)
  listey.append(x)

plot(listex, listey, '.k')
xlim(0, 50)
ylim(0, 50)
show()
```



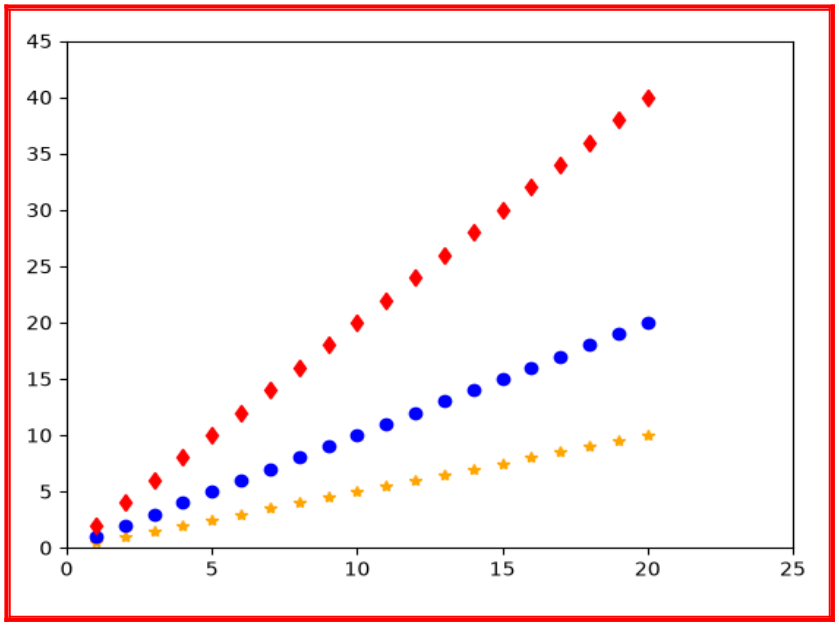
Exercice 7 :

```
from pylab import *

lx = []
ly1 = []
ly2 = []
ly3 = []

for x in range(1, 21):
  lx.append(x)
  ly1.append(0.5 * x)
  ly2.append(1 * x)
  ly3.append(2 * x)

plot(lx, ly1, '*', color='orange')
plot(lx, ly2, 'ob')
plot(lx, ly3, 'dr')
xlim(0, 25)
ylim(0, 45)
show()
```

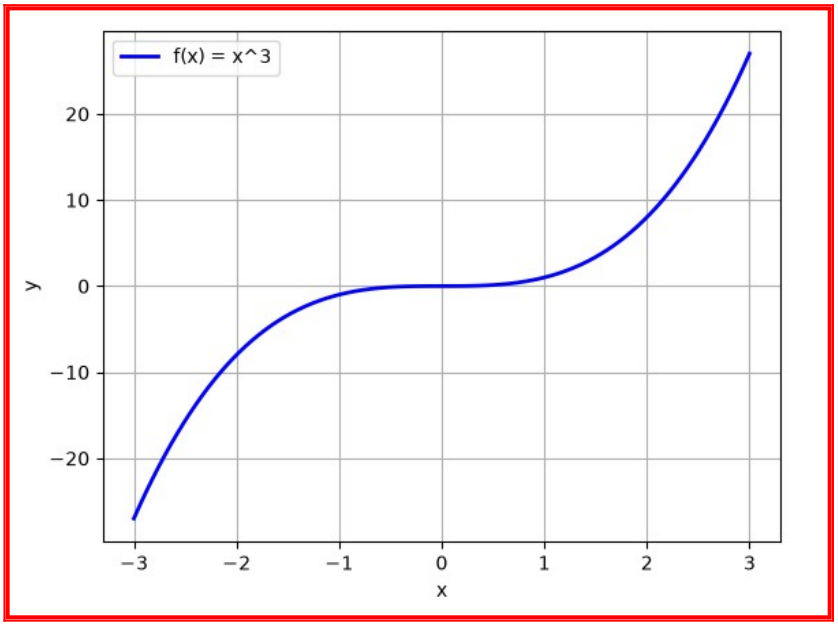


Exercice 8 :

1- Tracer la courbe représentative de la fonction cube sur l'intervalle [– 3 ; 3] avec 100 points.

```
from pylab import *
import numpy as np
x = np.linspace(-3, 3, 100)
y = x**3
plot(x, y, 'b-', linewidth=2, label='f(x) = x^3')
xlabel('x')
ylabel('y')
grid(True)
legend()

show()
```



2- Tracer la courbe de la fonction inverse sur l'intervalle [– 5 ; 5] avec 500 points.

```
from pylab import *
import numpy as np

x1 = np.linspace(-5, -0.01, 250)
x2 = np.linspace(0.01, 5, 250)

y1 = 1 / x1
y2 = 1 / x2

plot(x1, y1, 'r-', linewidth=2, label='f(x) = 1/x')
plot(x2, y2, 'r-', linewidth=2)

xlabel('x')
ylabel('y')
xlim(-6, 6)
ylim(-10, 10)
grid(True)
legend()

show()
```

